

Experiment 5:

Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

Python Code:

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
X = np.array([[1,2], [2,3], [3,4], [4,5], [5,6], [6,7]])
y = np.array([0, 0, 0, 1, 1, 1])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
new_data = np.array([[5,6]])
predicted_class = knn.predict(new_data)
print("Actual values:", y_test)
print("Predicted values:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("New input:", new_data[0])
print("Predicted class:", predicted_class[0])
```

Output:

```
Actual values: [0 0]
Predicted values: [1 1]
Accuracy: 0.0
New input: [5 6]
Predicted class: 1
```